Uni. Roll No:-.....

SET – B

GLA University, Mathura

Course: -B.Tech. (CSE) II – Year, II – Mid Term (Even Sem.) Examination, 2012 – 13

Subject: - Database Management System (ITE 221)

Time:-90 Minutes **M.M:-20**

Notes:-

1. Answer all questions from Group A (which is) compulsory, Any Two from Group B and Any Two from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group – A (All Questions are Compulsory) **08 X 0.5 = 04**

I A query within a query is called as –

- New Query
- Subquery
- Parent Query
- Master query

II The logical tables of data, extracted from existing tables are known as –

- Views
- Records
- Queries
- None of these

III Which of the following is the result of a SELECT statement –

- Table
- View
- Index
- Trigger

IV The _____ clause specifies a summary query –

- ORDER BY
- GROUP BY
- WHERE
- HAVING

V The process of decomposition of a table is known as –

- Generalization
- Specialization
- Normalization
- Compilation

VI After normalization, the original table can be obtained by which operation –

- Cascade
- Delete
- Join
- Selection

VII A table which is in 2NF, must be in of –

- 2NF
- 1NF
- 3NF
- 4NF

VIII The process of normalization is –

- Non – reversible
- Reversible
- Iterative
- Recursive

Group – B (Answer any TWO) **03 X 2 = 06**

I Write a PL/SQL code for finding volume of the cylinder.

II Define Partial functional, Transitive and Trivial dependency with the help of suitable examples.

III Explain how a Group by clause works in SQL also differentiate between WHERE and HAVING clause?

Group – C (Answer any TWO) **05 X 2 = 10**

I Consider the universal relation R = {A, B, C, D, E, F, G, H, I, J} and the set of functional dependencies are = {{A, B → C}, {A → D, E}, {B → F}, {F → G, H}, {D → I, J}}. What is the key for R? Decompose R into 2NF and then 3NF.

II Consider the following database : EMPLOYEE (emp_no, emp_nm, designation, dob, doj, salary, dno), DEPARTMENT (dno, d_nm, d_loc), ALLOWANCE (designation, special_allowance, commission)
Now write the SQL syntax of following queries :

1. Find employee number & employee name of employees who are born on 08 – Sep – 81.
2. Find the employee names who are working in '*Information Security*' department.
3. Display the total salaries in each department wise.
4. Find the total salary and average salary of each employee designation wise.
5. Give all employees a 10% raise in their salary.

III Explain lossy join decomposition and loss less join decomposition with the help of suitable example.

Uni. Roll No:-.....

SET-B

GLA University, Mathura

Course: -B.Tech. II-Year, II-Mid Term (Even Sem.) Examination, 2012-13

Subject: - Operating System (CSE-221)

Time:-90 Minutes **M.M:-20**

Notes:-

1. Answer all questions from Group A (which is) **compulsory**, **Any Two** from Group B and **Any Two** from Group C.
2. All questions of the particular group should be answered collectively at one place. All Parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group-A (All questions are compulsory) 08X 0.5=04

1. _____ is a technique of improving the priority of process waiting in Queue for CPU allocation.

- Starvation
- Aging
- Increment
- Critical Section

2. To avoid race condition, the maximum number of processes that may be simultaneously inside the critical section is

- two
- zero
- one
- more than two

3. The protocol where before proceeding with its execution, each process must acquire all the resources it needs is used in

- Mutual Exclusion
- Circular Wait
- Hold and Wait
- No Pre-emption

4. An operating system contains 3 user processes each requiring 2 units of resource R .The minimum number of units of R such that no deadlocks will ever arise is

- 3
- 5
- 6
- 4

5. A critical section is a program segment

- which should run in a certain specified amount of time.
- which avoids deadlocks.
- where shared resources are accessed.
- which must be enclosed by a pair of semaphore operations, P and V.

6. Which scheduling policy is most suitable for a time-shared operating system?

- Shortest-Job First
- FCFS
- Elevator
- Round-Robin

7. Let S and Q be two semaphores initialized to 1, where P0 and P1 processes the following statements wait(S);wait(Q); ---; signal(S);signal(Q) and wait(Q); wait(S);---;signal(Q);signal(S); respectively. The above situation depicts a _____ .

- Semaphore
- Interrupt
- Signal
- Deadlock

8. Which of the following approaches do not require knowledge of the system state?

- Deadlock Detection
- Deadlock Avoidance
- Deadlock Prevention
- None of the above

Group-B (Answer any Two) 3x2=6

Q1. What is a race condition? What are all the conditions that should hold for its good solution? Explain any one hardware implementation of this problem.

Q2. What do you mean by RAG? Construct a RAG using 4 processes and 4 resource typs such that
(iii) The graph has cycle and processes are deadlocked.
(iv) The graph has cycle but processes are not deadlocked.

Q3. Explain in detail what is dining philosopher problem.

Group-C (Answer any Two) 5x2=10

Q1. Consider the following set of processes:-

PROCESS	BURST TIME(msec)	ARRIVAL TIME(msec)
A	75	0
B	40	10
C	25	10
D	20	80
E	45	85

Answer the following questions:

d) Draw Gantt chart using FCFS, SRTF and Round-Robin Scheduling with time quantum 15msec.
e) What is the Turnaround time of processes of FCFS and SRTF CPU Scheduling?
f) What is the average waiting time of processes using Round-Robin scheduling

Q.2. Consider the following system snapshot using data structures in the Banker's algorithm, with resources A, B, C, and D, and process P0 to P4:

	MAX				ALLOCATION				AVAILABLE			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1	3	2	1	1
P1	1	7	5	0	1	1	0	0				
P2	2	3	5	6	1	2	5	4				
P3	1	6	5	3	0	6	3	3				
P4	1	6	5	6	0	2	1	2				

Using Banker's algorithm, answer the following questions.
(a) How many resources of type A, B, C, and D are there?
(b) What are the contents of the Need matrix?
(c) Is the system in a safe state?
(d) If a request from process P4 arrives for additional resources of (1, 2, 0, 0,), can the Banker's algorithm grant the request immediately?

Q.3. What are semaphores? Give the solution of Reader-Writer's Problem using semaphore.

GLA University, Mathura **SET-B**

Second mid Term Examination, 2012-13

Course: - B.Tech **II-Year, IV-Sem (Sec-A,B,C,D)**

Subject: - Introduction to Microprocessor (CSE222) **Uni. Roll No:-**

Time:-90 Minutes **Total Marks:-20**

Notes:-

1) **Answer all questions, Section A is compulsory, Any two from Section B and Any two from Section C.**
2) **All parts of a question (a, b, etc.) should be answered at one place.**
3) **Answer should be brief and to-the-point and be supplemented with neat sketches.**
4) **Any missing or wrong data may be assumed suitably giving proper justification.**
5) **Figures on the right-hand side margin indicate full marks.**

Part A

Q1) Attempt following questions. **[0.5x08=04]**

i) Which provide a mechanism to establish a link between the microprocessor and I/O device?
- Input & output
- Output interface
- Input interface
- None of these

ii) The processor of knowing the status of device and transferring the data with matching speeds is called
- Ports
- Peripherals
- Handshaking
- None of theses

iii) In a 16-bit microprocessor, words are stored in two consecutive memory locations. The entire word can be read in one operation provided the first
- Memory location is odd.
- Word is odd
- Word is even
- Memory add is even

iv) In 8086 microprocessor one of the following instructions is executed before an arithmetic operation
- DAS
- AAD
- AAM
- DAA

v) In 8086 the over flow flag is set when
- Signed numbers go out of their range after an arithmetic operation
- The sum is more than 16 bit
- Carry & Sign flag are set.
- Zero flag is set

vi) What is Op-code?
- The value in which an operation acts upon
- The instruction that is to be executed
- A mnemonic that defines a data size
- The compiled assembly code

vii) What are the names of the four segment registers?
- Stack, Data, Base, Counter
- Stack, Index, Extra, Code
- Data, Index, Code, Stack
- Stack, Extra, Code, Data

viii) The 8086/8088 used two processing units which were known as
- Bus Unit and Execution IU
- Segment and Offset Units
- Left and Right Units
- Bus Interface Unit and EU

Part B

Q2) Attempt any two of the following. **[3x2=06]**

a) Explain the maximum memory addressing and I/O addressing capability of 8086 with neat sketches?

b) List and explain the internal registers of 8086 microprocessor with their abbreviations.

c) Calculate the delay in the following loop, assuming the system clock period is o.33 μs:

Label	Mnemonics	8085 T-States
	LXI B, 12FFH	10
DELAY:	DCX B	6
	XTHL	16
	XTHL	16
	NOP	4
	NOP	4
	MOV A,C	4
	ORA B	4
	JNZ DELAY	10/7

Part C

Q3) Attempt any two of the following. **[5x2=10]**

a) List the 8086 addressing modes, and given as example of each. Also explain the flags of 8086?

b) Specify the number of times the following loops are executed. Also explain the flow of execution.

i)
```
        MVI  A, 17H
LOOP:   ORA  A
        RAL
        JNC  LOOP
```

ii)
```
        MVI  A, 17H
LOOP:   RAL
        ORA  A
        JNC  LOOP
```

iii)
```
        LXI  B, 1000H
LOOP:   DCX B
        NOP
        JNZ  LOOP
```

c) With the help of block schematic diagram explain the operation of 8086 processor.

GLA University, Mathura **SET A**

II-Mid Term Examination, 2012-13

Course: - B.Tech. **II-Year, IV-Sem.**

Subject: - Computer Graphics (CSE 223) **Uni. Roll No:-**

Time:-90 Minutes **Total Marks:-20**

Notes:-

1) Answer all questions, Section A is **compulsory, Any Two** from Section B and **Any Two** from Section C.
2) All parts of a question (I, II, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

SECTION-A

Objective Questions: (0.5*8 = 04)

I. In perspective projection, the center of projection is at

- infinity
- finite distance
- x-axis
- y-axis

II. In superellipsoid value of S1,S2 are equal to 1, then the form of super ellipsoid will be:

- ellipse
- ellipsoid
- torus
- paraboloid

III. A projection in which all three foreshortening factors are kept equal is called as:

- Isometric projection
- Diametric projection
- Trimetric projection
- all of these

IV. If $S_x=S_y=0$ then it means:

- Uniform reduction
- Object collapses
- Uniform scaling
- None of these

V. The join points of the curve are also known as:

- Plots
- Knots
- Clots
- Dots

VI. The degree of a Bezier curve segment is determined by the

- Number of control points to be fitted with that curve segment
- Interpolation of control points
- Geometric continuity of the curve
- None of these

VII. In B-SPline curve for n+1 control points, the curve is described with blending functions:

- n-1
- n+1
- n
- n^2

VIII. The transformation in which an object is changed relative to stationary coordinate system is called as:

- Coordinate transformation
- Geometric transformation
- Translation
- Reflection

SECTION B (Attempt any two)

I. Explain Bezier curves with example. Write the properties of Bezier curves. (1.5+1.5)

II. Prove that simultaneous shearing in both directions (x and y) is not equal to the composition of pure shear along x-axis followed by pure shear along y-axis. (3)

III. Specify the significance of continuity conditions. Discuss parametric continuity conditions and differentiate it with geometric continuity condition. (0.5+1.5+1)

SECTION C (Attempt any two)

I. What is Projection? Differentiate Parallel and Perspective projection. A unit cube is projected on to the xy plane find out the coordinate of the projected image using perspective projection, if center of projection is (0, 0, 10). (1+1+3)

II. Rotate a triangle at **A (0, 0), B (1,1), C (5,2)** by 45^0 about the point P (-1,-1). (5)

III. Find the coordinates of a pyramid whose coordinates are A(0,0,0), B(1,0,0), C(0,1,0) and D(0,0,1) after mirror reflection with respect to the plane passing through the origin, and having the normal vector N= i+j+j. (5)

Uni. Roll No:-.....

SET-A

GLA University, Mathura

Course: -B.Tech. II-Year, II-Mid Term (Even Sem.) Examination, 2012-13

Subject: - Computer Network

Time:-90 Minutes **M.M:-20**

Notes:-

1. Answer all questions from Group A (which is) **compulsory, Any Two** from Group B and **Any Two** from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group-A (All questions are compulsory) 08X 0.5=04

Q1. Using CRC method for error detection, the remainder at the receiver should be _____to detect no error.

- **Non-Zero**
- **Zero**
- **The quotient**
- **Equal to the remainder at the sender**

Q2. What is FDDI ?

- **Bus Based network**
- **Ring network**
- **Star network**
- **Mesh network**

Q3. In class C IP addressing method, the range of values that the first octet can have is

- **192-255**
- **1-126**
- **192-223**
- **128-226**

Q4. Which addresses of the following is the loopback address?

- **127.0.0.0**
- **127.0.0.1**
- **0.0.0.1**
- **255.255.255.255**

Q5. Menchester encodisng is used in

- **IEEE 802.1**
- **IEEE 802.4**
- **IEEE 802.2**
- **IEEE 802.3**

Q6. Which of the following is the address of the router?

- **IP address**
- **TCP address**
- **Subnet Mask**
- **Default Gateway**

Q7. In slotted ALOHA, a maximum throughput of _____ percent can be achieved, while it is _____ percentage for pure ALOHA.

Q8. _____ is abbreviated as CSMA/CD and is also known as ._____

Group-B (Answer any Two) 3x2=6

Q1.Explain the following ARQ techniques in detail

a) Stop and Wait ARQ b) Selective Repeat ARQ

Q2. Sixteen-bit messages are transmitted using a Hamming code. How many check bits are needed to ensure that the receiver can detect and correct single bit errors? Show the bit pattern transmitted for the message 1101001100110101. Assume that even parity is used in the Hamming code.

Q3.Generate the CRC for a frame 1101011011 using the generator $G(x) = x^4 + x + 1$.show the results in each steps of the generation.

Group-C (Answer any Two) 5x2=10

Q1.Explain the working of IEEE 802.3,IEEE802.4,IEEE 802.5 with proper diagrams for each.

Q2. A large population of ALOHA users manages to generate 50 requests/sec, including both originals and retransmissions. Time is slotted in units of 40 msec.

(a) What is the chance of success on the first attempt?

(b) What is the probability of exactly k collisions and then a success?

(c) What is the expected number of transmission attempts needed?

Q3.A router has the following (CIDR) entries in its routing table:

Address/mask	**Next hop**
135.46.56.0/22	Interface 0
135.46.60.0/22	Interface 1
192.53.40.0/23	Router 1
Default	Router 2

For each of the following IP addresses, what does the router do if a packet with that address arrives?

(a) 135.46.63.10 (b) 135.46.57.14 (c) 135.46.52.2 (d) 192.53.40.7 (e) 192.53.56.7

Uni. Roll No:-.....

GLA University, Mathura

SET-C

Course : B.Tech.,II-Year, II-Mid Term (Even Sem.) Examination,2012-13

Subject : Environmental Studies (CEE-201)

Time: 90 Minutes **M.M : 20**

Note:

1. Answer all questions from Group A (which is) **compulsory**, **Any Two** from Group B and **Any Two** from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group- A

Attempt All Questions.

0.5X8= 4 Marks

I. According to current Census Report (2010), the population of India is-

• 6.8 billion • 1.2 billion • 3.4 billion • 2.3 billion

II. Agricultural run-off containing nitrogen and phosphorous into water bodies causes-

• Salinization • Eutrophication • Bio-magnification • All of these

III. Which is not an example of bio-fuel in following-

• Coal • Natural gas • Bio-gas • Petroleum

IV. The permissible noise quality standard for fire crackers is-

• 175 dB • 125 dB • 250 dB • 180 dB

V. *Ex- citu* conservation of bio-diversity includes-

• Gene banks • National parks • Wild life sanctuaries • Forests

VI. The atmospheric gas that has the highest potential of causing green house effect and global warming from anthropogenic sources is-

• O_3 • CO_2 • CH_4 • N_2O

VII. The gas leaked in Bhopal Gas Tragedy was-

• CO • MIC • NH_3 • PH_3

VIII. The main source of carbon in atmosphere is -

• Atmospheric CO_2 • Photosynthesis • Respiration •Burning of fossil fuels

Group- B

Attempt Any Two Questions.

3X2= 6 Marks

I. Define thermal and photo-voltaic conversions of solar energy with any two applications of each.

II. How will you dispose religious-cultural botanical waste to prevent Yamuna River pollution.

III. Explain "acid rain" by discussing the case study of Taj Mahal at Agra.

Group- C

Attempt Any Two Questions.

5X2= 10 Marks

I. What is ozone hole? How it is formed ? Suggest suitable control to protect ozone layer.

II. What is air pollution ? Classify air pollutants with suitable examples. Give any two approaches to prevent air pollution.

III. (a) Compare hydro-power and fossil fuel based thermal power. Suggest any four non-conventional sources for power generation also.

(b) Enumerate various effects of industrialization on environment.